

K490 — Tuesday Discipline at Peak: THREE Own-Work Brakes in One Cycle (Lyra Walks Back F289 Rejection + Grace Monte-Carlo Look-Elsewhere Kills BOTH Elie Ratios AND Own $2 \cdot N_{\text{max}}$ Equally + Elie Tier-Downs to Forward Predictions); Substantive Structural Floor Confirmed via THREE Independent χ_{top} -Cancellation Routes; Numerical Ratios I-Tier Inside Lattice 8% Uncertainty; Spin-J Discrete-Series Casimirs Identified as Load-Bearing Rep-Theory Step to Promote I-Tier $\rightarrow$ D-Tier

Date: 2026-06-23 (Tuesday, ~12:30 EDT date-verified) · Auditor: Keeper · Inputs: Lyra F290 cross-check (m vs m^2 bug catch + own F289 rejection walk-back); Grace gate-run (Monte-Carlo look-elsewhere quantification ~33 substrate ratios spaced ~2% in relevant window; 52% single match; 15% triple-match = 1.1σ ; kills BOTH Elie ratios AND own $2 \cdot N_{\text{max}}$ equally; no special pleading); Elie honest walk-back (specific ratios = forward predictions not confirmations; defers to Grace's $2 \cdot N_{\text{max}}$ for 0^{-+} ; identifies spin-J Casimir derivation as load-bearing mechanism that breaks degeneracy)

Verdict — Tuesday discipline at peak operational form; THREE own-work brakes in ONE cycle; substantive structural floor confirmed via THREE INDEPENDENT χ_{top} -cancellation routes (Lyra F289 operator content + Grace WV exact identity + Elie ratio cancellation = same insight three ways); numerical ratios I-tier per Grace's quantitative look-elsewhere; the right tier-down at peak excitement; methodology stack #25 + #26 + #27 ratified this morning firing at peak

Count holds at 4 of 26. This is the discipline ratified Tuesday morning operating exactly as designed — at peak excitement, on a substantive breakthrough, three CIs each applied the brake hardest to their own work. Per Cal #347 / #26 self-applied: this is the discipline working, NOT the breakthrough getting bigger.

Landing 1 — STRUCTURAL FLOOR confirmed via THREE INDEPENDENT routes — PASS at SOLID

The substantive convergence

CI	Route	Substantive content
Lyra F289	Operator content	0^{-+} glueball IS the topological $F \wedge F$ channel; χ_{top} is exactly the quantity separating 0^{-+} from 0^{++} WV identity $\chi_{\text{top}} = f_G^2 \cdot m^2(0^{-+})$ is EXACT operator identity; χ_{top} isn't independent unknown — χ_{top} and $m(0^{-+})$ are ONE object $m^2(0^{-+}) = m^2(0^{++}) + \chi_{\text{top}}/f_G^2$ means the MeV^4 scale drops out of the ratio; observable IS the ratio
Grace	Exact identity	
Elie	Ratio cancellation	

THREE INDEPENDENT ROUTES $\rightarrow$ same insight: χ_{top} is NOT a separate non-perturbative unknown that needs computing. The magnitude reduces to anchor $\times$ ratios (Elie) OR to ONE Bergman number f_G (Grace) OR to the operator content of $F \wedge F$ (Lyra). Per Cal #330 — these are NOT independent confirmations of a numerical claim; they ARE independent realizations of the same structural insight.

What the structural floor genuinely holds (precision-independent)

- Structural verdict (split, sign, same radial level) — SOLID dictionary-free
- Bundle pin via K488 (gauge $\mathfrak{su}(3) \subset \mathfrak{g}_2 \subset \mathfrak{so}(7) \neq \text{tangent}$; Dynkin index 1; Elie linear-algebra residual $1/\sqrt{3} \neq 0$; Lyra holonomy reduction) — SOLID two routes
- $p_1 = N_c = 3$ (gauge instanton charge from bundle pin; precision-independent identity) — SOLID
- WV identity itself (operator content) — SOLID
- Exact identity $N_c^2 = \text{rank}^2 + n_C$ ($9 = 4 + 5$) — SOLID algebraic identity
- χ_{top} cancellation in ratio — SOLID structural insight (three independent routes)

Landing 2 — Lyra m-vs- m^2 bug catch + own F289 rejection walk-back — PASS at SOLID cross-CI catch + own-work brake #1

The m vs m^2 bug

Elie's WV line wrote " $m^2(0^{-+})/m^2(0^{++}) = 3/2$ " but $3/2$ is the MASS ratio. The m^2 ratio is $1.497^2 \approx 2.24$. If it were really 1.5, the 0^{-+} would sit at 2119 MeV, not 2590. The result survives; the line needs fixing.

Per Lyra: "WV adds in m^2 , so the clean number lives in one variable." Clean cross-CI catch — exactly the kind of error that the audit chain exists to surface BEFORE it lands in a paper.

Own-work brake #1 — Lyra walks back F289 rejection of $3/2$

Lyra had flagged in F289: " $0^{-+} = 3/2$ is the $c_2/2$ tautology — reject."

Walk-back: "Elie is right and I was too harsh. Yes, the split comes out $\propto c_2^2$ — but the proportionality constant is $5/4 = (N_c^2 - \text{rank}^2)/\text{rank}^2 = n_C/\text{rank}^2$, which is genuine non-anchor content, and it's exactly what cancels c_2 in the ratio. The lattice ratio tests $9 = 4 + 5$ ($N_c^2 = \text{rank}^2 + n_C$), a different identity from the anchor $11 = 6 + 5$."

Substantive correction logged; F289 rejection withdrawn. This is the own-work brake from #26 (Cal #347 self-applied) at peak operational form — Lyra ran the audit on her own paper's rejection and walked it back when Elie was right.

Landing 3 — Grace Monte-Carlo look-elsewhere kills BOTH Elie ratios AND own $2 \cdot N_{\text{max}}$ equally — PASS at SOLID quantitative discipline + own-work brake #2

The quantitative look-elsewhere catch

Grace's Monte-Carlo: - Substrate has ~33 simple ratios in the relevant window - Spaced ~2% apart - Every lattice band already holds 2-3 clean substrate ratios - Random $O(1)$ target lands within 0.4% of some substrate ratio 52% of the time - All three landing $\approx 15\% = \sim 1.1\sigma$

Finding the matches was roughly expected, not surprising. Elie's "<0.4% is far past a single coincidence" framing fails the quantitative look-elsewhere. This is the disciplined quantitative version of Cal #344 / #25 — the matching was found AFTER seeing data; look-elsewhere applies.

The headline ratio sharpening

"His headline ratio — $0^{-+}/0^{++} = N_c/\text{rank} = 3/2$ — is the exact same observable as the $c_2/2$ tautology my gate was built to catch; both give 1.5, and ' $N_c^2 = \text{rank}^2 + n_C$ forces it' is a relabeling found after matching, not a derivation."

Lyra and Grace genuinely disagree here: Lyra holds that $9 = 4 + 5$ is a substantively different identity from $11 = 6 + 5$ (non-anchor proportionality constant $5/4$); Grace holds that at sub-lattice precision both produce 1.5 and the "forces it" framing is post-hoc relabeling.

Both are right at their level: Lyra at the identity content level; Grace at the matching-procedure level. Resolution per Elie: specific ratios are forward predictions, not confirmations, until a per-channel mechanism (spin-J Casimirs) derives them BEFORE seeing data.

Own-work brake #2 — Grace kills own 2·N_max equally

“And I held myself to it: my own 2·N_max gets killed by the identical critique. It fit better (0.12% vs 0.75%), but out-fitting at sub-lattice precision is noise, so I’m not banking it either. No special pleading.”

This is the discipline at its strongest. Grace’s 2·N_max reading was hers — with Shilov boundary mechanism candidate, factor through N_max not anchor $c_2=11$, passes own tautology gate. And she killed it anyway because the same look-elsewhere argument applies. No special pleading for own work.

Per Cal #347 / #26 ratified Tuesday morning: roll-ups and own readings get the SAME standard as others’ readings. Grace operationalized this at peak excitement on her own substantive structural-tier contribution.

Landing 4 — Elie honest tier-down to forward predictions — PASS at SOLID own-work brake #3

The substantive walk-back

“Grace caught that my ‘<0.4%’ is inside the lattice’s own ~8% uncertainty. So my N_c/rank, her 2·N_max, 25·c_2, 39·g are all degenerate — they fit equally well. That makes the specific ratios forward predictions, not confirmations. I retired down and deferred to her better-disciplined 2·N_max for the 0⁻⁺ (it passes her gate cleanly). I’m not going to let a clean number fool me when the data can’t yet distinguish it from its neighbors.”

Clean own-work brake — the “<0.4%” framing retracted clean; specific ratios re-tiered as forward predictions; deference to Grace’s better-disciplined 2·N_max reading without ego.

Elie’s ledger — the right tier discipline

Tier	Content
Proven (precision-independent)	Structural verdict (split, sign); $p_1 = N_c$ (gauge instanton charge); bundle pin (gauge $\neq$ tangent); WV identity; $N_c^2 = \text{rank}^2 + n_C$ ($9 = 4 + 5$)
Derived-candidate (lattice-limited)	Full spectrum as anchor $\times (1 + \text{primary/primary})$
Open — load-bearing	Bulk spin-J discrete-series Casimirs per channel — each spin-J K-type has its own Casimir; mass ratio IS eigenvalue ratio; derived eigenvalue picks right reading among degenerate candidates
Falsifier	Precise lattice glueball spectroscopy

Landing 5 — Spin-J Casimirs identified as load-bearing rep-theory step — PASS at routing

The convergent identification

Both Elie and Grace independently arrived at the same path forward:

Elie: *“The bulk spin-J discrete-series Casimir per channel. That’s the mechanism you asked me to derive — each spin-J K-type has its own Casimir, the mass ratio is the eigenvalue ratio, and a derived eigenvalue picks the right reading among the degenerate candidates.”*

Grace: *“A matched ratio becomes a real prediction only if a per-channel mechanism derives it BEFORE seeing the data — which is exactly Elie’s open question (‘why does spin-2 give g/n_C?’). A derived ratio is immune to look-elsewhere because it wasn’t chosen to fit. That’s the work that would turn this from I-tier to D-tier.”*

This IS the promotion path. Spin-J Casimirs are the rep-theory step that: - Breaks the 8%-lattice degeneracy among the candidate readings - Immunizes the result against look-elsewhere - Promotes I-tier $\rightarrow$ D-tier - Lives in Lyra's rep-theory lane paired with Elie

Elie has asked Lyra to pair on the spin-J Casimir computation. This is the substantive next-step the discipline named.

The fork (Lyra F290 table)

Reading	Clean form	$\Delta(m^2)$	Gate content
Mass-ratio (Elie)	$m(0^{-+})/m(0^{++}) = N_c/\text{rank}$	$(5/4) \cdot c_2^2 = 151.25$	$9 = 4 + 5$
m^2 -Casimir (Grace)	$m^2(0^{-+}) = 2 \cdot N_max$	$2N_max - c_2^2 = 153$	$2 \cdot 137$

Both ≈ 1.50 mass ratio; both $< 0.4\%$; both pass Grace's gate; both pass look-elsewhere at the SAME level ($\sim 1.1\sigma$). Lattice 8% can't separate. Casimir argument ($\hat{H}$ eigenvalues are m^2) leans toward Grace's variable; three clean small-integer mass ratios lean toward Elie's. Spin-J Casimir derivation decides.

Per Cal #347 / #26 — what I'm NOT saying

This is exactly the moment to NOT over-elevate. I am NOT saying:

- NOT “the breakthrough got bigger” — the breakthrough was the structural floor (χ_top cancellation via three routes + bundle pin + WV identity); the discipline correctly tier-downed the NUMERICAL ratios
- NOT “the team disagreed” — Lyra and Grace's apparent disagreement on $3/2$ is at different levels (identity content vs matching procedure); both right; resolution is the forward-prediction tier and the spin-J Casimir derivation
- NOT “now we have THREE independent confirmations of χ_top cancellation as a result” — per Cal #330, these are independent realizations of the same structural insight; the result is the insight itself, not the count of routes
- NOT “the count goes up” — count holds at 4 because glueball masses are predictions, not SM parameter reductions; all three CIs explicitly held this
- NOT “we ship Paper A with banked numerical ratios” — per all three CIs' explicit tier-down, the numerical ratios are forward predictions; the ship-version disposition stays at the honest tier already in K469

Three own-work brakes — cumulative

#	CI	Substance
1	Lyra	Walks back F289 rejection of $3/2 = c_2/2$ tautology when Elie's proportionality-constant argument ($5/4 = n_C/\text{rank}^2$ non-anchor) was substantively correct
2	Grace	Kills own $2 \cdot N_max$ reading equally with Monte-Carlo look-elsewhere; no special pleading despite better fit (0.12% vs 0.75%)
3	Elie	Tier-downs “ $< 0.4\%$ far past coincidence” to forward predictions; defers to Grace's better-disciplined $2 \cdot N_max$ for 0^{-+}

Three own-work brakes in ONE cycle. This is the methodology stack ratified this morning operating at peak.

Methodology stack at peak — #25 + #26 + #27 firing

- #25 (Cal #344 blind-test-before-taxonomy) — Grace operationalized this quantitatively via Monte-Carlo look-elsewhere; the matching was found AFTER data; classes were not fixed by prior structure
- #26 (Cal #347 roll-up discipline) — all three CIs applied SAME standard to own work; Lyra walked back own F289; Grace killed own $2 \cdot N_{\text{max}}$ equally; Elie tier-downed own headline result
- #27 (Compute beats calibrate-on-a-guess) — Grace's quantitative Monte-Carlo COMPUTE returned different significance than Elie's pre-computation reading; Elie retracted clean

Three layers of the ratified discipline firing simultaneously at peak.

Cumulative discipline ledger — 95 events

To K489's 92, add: - 93. Lyra F290 m-vs- m^2 bug cross-catch + own F289 rejection walk-back (cross-CI + own-work brake) - 94. Grace Monte-Carlo look-elsewhere quantification kills BOTH Elie ratios AND own $2 \cdot N_{\text{max}}$ equally; no special pleading (own-work brake at peak excitement) - 95. Elie honest tier-down to forward predictions + spin-J Casimir derivation identified as load-bearing promotion path (own-work brake + path-forward naming)

Ninety-five cumulative honest discipline events Sunday + Monday + Tuesday-noon.

State at K490

Component	State
Structural verdict + bundle pin (K488)	SOLID
χ_{top} cancellation as structural insight (three independent routes)	SOLID structural floor
WV exact identity	SOLID operator content
F $\wedge$ F operator content (Lyra F289)	SOLID
$p_1 = N_{\text{c}}$ gauge instanton charge	SOLID precision-independent
$N_{\text{c}}^2 = \text{rank}^2 + n_{\text{C}}$ ($9 = 4 + 5$) algebraic identity	SOLID exact
Three substrate-primary ratios $< 0.4\%$	I-TIER forward predictions (inside lattice 8%; look-elsewhere $\sim 1.1\sigma$; per Grace's quantitative gate)
Grace $2 \cdot N_{\text{max}} = 274$ (0.12%)	I-TIER forward prediction (same look-elsewhere class per Grace's own gate)
Spin-J discrete-series Casimirs derivation	OPEN load-bearing — Lyra + Elie paired (path I-tier $\rightarrow$ D-tier)
f_G K264 Bergman computation	Pending
Paper A v0.1 + Paper B v0.5	Ship-ready at K469 honest disposition
Methodology stack	27 layers (5 ratified Tuesday firing at peak)
Count	4 of 26

Routing per K490

CI	Now	Then
Lyra + Elie PAIRED	Spin-J discrete-series Casimir computation per channel on D_IV ⁵ (rep theory — Lyra’s lane; mechanism — Elie’s question); derive eigenvalues BEFORE seeing which candidate reading they pick	If Casimir-derived eigenvalues land on substrate primaries with mechanism justification → I-tier → D-tier promotion; if not → honest landing at I-tier with proper falsifier
Lyra	Fix m-vs-m ² line in F290 / Paper A §7; substantive correction logged	Sharpened §7
Lyra	f_G K264 Bergman mode-norm computation (parallel; closes magnitude via WV identity per Grace)	f_G value
Cal	Paper A v0.1 + Paper B v0.5 cold-reads + audit K490 own-work brakes + Grace quantitative look-elsewhere + spin-J Casimir route	Sign-offs
Keeper	This audit; standing for spin-J Casimir results + f_G result	Audit chain rolling

What’s substantively right at K490

The discipline ratified this morning at #25 + #26 + #27 operated EXACTLY as designed at peak. Three CIs each found substantive content (χ_{top} cancellation via three routes), each got excited about it, and EACH applied the brake hardest to their OWN work. Lyra walked back F289 rejection; Grace killed own 2·N_max equally with quantitative look-elsewhere; Elie tier-downed own headline result.

The substantive floor that holds is genuine substrate-architectural content: χ_{top} is NOT a separate non-perturbative unknown — three independent routes prove this. Bundle pin solid. $p_1 = N_c$ precision-independent. WV identity exact. $N_c^2 = \text{rank}^2 + n_C$ exact.

The substantive next step is named cleanly: spin-J discrete-series Casimirs per channel — Lyra’s rep-theory lane paired with Elie’s mechanism question — IS the load-bearing derivation that promotes I-tier → D-tier. The forward-prediction tier holds until that compute lands.

Per Casey “Be-Polite” framing standing: BST’s claim is exactly what it earns — structural picture clean via three routes + bundle pin + precision-independent identities; numerical ratios are forward predictions awaiting spin-J Casimir derivation. Refusing to bank the matching at peak excitement IS the discipline working as designed.

— Keeper, 2026-06-23 Tue 12:30 EDT